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Toward optimal feature and time segment selection by divergence method for EEG signals classification

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ABSTRACT

Feature selection plays an important role in the field of EEG signals based motor imagery pattern classification. It is a process that aims to select an optimal feature subset from the original set. Two significant advantages involved are: lowering the computational burden so as to speed up the learning procedure and removing redundant and irrelevant features so as to improve the classification performance. Therefore, feature selection is widely employed in the classification of EEG signals in practical brain-computer interface systems. In this paper, we present a novel statistical model to select the optimal feature subset based on the Kullback-Leibler divergence measure, and automatically select the optimal subject-specific time segment. The proposed method comprises four successive stages: a broad frequency band filtering and common spatial pattern enhancement as preprocessing, features extraction by autoregressive model and log-variance, the Kullback-Leibler divergence based optimal feature and time segment selection and linear discriminate analysis classification. More importantly, this paper provides a potential framework for combining other feature extraction models and classification algorithms with the proposed method for EEG signals classification. Experiments on single-trial EEG signals from two public competition datasets not only demonstrate that the proposed method is effective in selecting discriminative features and time segment, but also show that the proposed method yields relatively better classification results in comparison with other competitive methods.

1. Introduction

Brain-computer interface systems (BCIs) are devised for the people to control devices or communicate with computers via their brain [1]. It aims to build a non-muscular output pathway to translate the brain activities into discriminative control commands associated with different electroencephalography (EEG) signals based motor imagery patterns [2]. It has proven that the potential changes recorded as EEG signals in motor area cortex either by imaging a real movement or executing it are similar to some extent [3]. Hence, the processing of EEG signals would supply a feasible solution to design such a BCI system. Generally, the acquisition of multichannel EEG signals, signal preprocessing, feature extraction and the EEG signals based motor imagery pattern classification are four basic steps in a classical BCI [4]. In most cases, existing researches pay more attention to the last two steps: feature extraction and pattern classification [5]. However, between the feature extraction and pattern classification procedures, feature selection is usually employed [6]. It should be noted that the feature selection procedure is crucial to improve the classification performance and the BCI practicability even though it is not indispensable. Besides, feature selection alone as well as applying feature selection to the motor imagery based BCI has been an active research field [4,6].

Feature selection, also termed feature reduction, could be defined as a procedure which forms a feature subset from the original feature set in order to optimally reduce the dimensionality of feature space according to a certain evaluation criterion [6]. Reducing the dimensionality of feature space could address two concerns. First, the training of a classifier may be computationally infeasible because of the large number of the involved features. Second, many features are redundant, irrelevant or noise dominated so that the inclusion of all features will be detrimental to the classification performance. It has proven that the use of feature selection can remove 98% irrelevant, redundant and noise distorted features without affecting the classification performance [7]. What is more, removing these features would greatly reduce the computational burden for training classifiers and improve the classification accuracy. Many experimental results demonstrate that higher classification accuracy

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could be achieved with selected features than with all the features [8]. However, in most cases, attempting a brute force selection of the best feature subset is intractable due to the curse of dimensionality. As the dimensionality of a domain expands, the number of features N increases and the search complexity of the best feature subset will increase exponentially. Hence, the feature selection related problems are shown to be non-deterministic polynomial (NP-hard) [9,10]. For instance, if we select the best subset from a bank of 50 features, the number of feature subsets is $2^{50}-1$ in which each feature could be selected or not. If we select 10 features from the original set, the number of subsets is $C_{50}^{10}=50!/10!(50-40)!$. It is in the order of 10^{10}, which is impractical in a real BCI design. Hence, in such a scenario, automatic or heuristic selection of the optimal feature subset is crucial to overcome the curse of dimensionality and deserves deep analysis [11]. Feature selection has been a fertile field in computer science, including statistical pattern recognition, machine learning and data mining [12]. Therefore, computer science would supply appropriate approaches to confronting the challenges from feature selection.

Feature selection algorithms could be divided into three categories [6,12,13]: embedded approaches, which perform feature selection embedded in the learning of optimal parameters; filter approaches, which rank the features based on their scores and evaluate the quality of selected features; wrapper approaches, which assess feature utility with respect to a predetermined classifier. Embedded approaches select the feature subset based on the intrinsic properties of parameter optimization, but they are unsuitable where there are significant interactions between relevant features [14]. Therefore, filter approaches and wrapper approaches are most widely used in feature selection because of their intuitive sense in selecting the optimal feature subset and their excellent classification performance. In filter approaches, a score that indicates the importance of a feature is assigned to each individual feature based on an independent evaluation criterion. Then the filter approaches select the top ranked features with high scores and discard the rest without the application of any classifier. Alternatively, the wrapper approaches greedily search for discriminative features by classification results obtained by a predetermined classifier. However, much more computational expense as well as poor general ability makes it impractical [13]. Because wrapper approaches consider the classification algorithm as a subroutine so that the sequential random division of feature subsets as well as superabundant cross-validations would be involved. Accordingly, filter approaches are more appealing and predominantly used because of their simplicity and efficiency. However, classical filter approaches estimate the goodness of each individual feature by only exploiting the intrinsic properties without considering the discriminative properties, which may lead to an undesired classification result.

In this paper, we present a novel feature subset selection method which ranks the original features and selects the top ranked features, aiming to maximize the discrimination performance for EEG signals based motor imagery pattern classification. Different from the classical filter approaches, the proposed method selects those features that offer maximum class separability instead of maximum information in terms of Kullback-Leibler (KL) divergence measure. Meanwhile, it evaluates the discriminative power of each individual feature without training a predetermined classifier explicitly. The proposed method has the ability to select the most discriminative features based on the maximum a posteriori estimation (MAP) rule, aiming to maximize the classification performance without involving any classification subroutine. Although the proposed method should be attributed to the filter approaches, it has the advantages involved in the wrapper approaches. After the feature selection procedure, the average KL-divergence of the selected features in each time segment is calculated, and the one with the largest value is chosen as the optimal time segment, which will be further clarified below. To accurately estimate the conditional probability within KL-divergence for each single feature, a Parzen window is employed without requiring permutation over all combination of features [15]. Besides, the feature extraction procedure includes autoregressive (AR) model and common spatial pattern logarithm (CSP-log) algorithm. It has proved that AR model and CSP-log algorithm are two powerful and efficient feature extraction and enhancement methods in biomedical information processing [4,14,16,17]. First, the frequency band filtered EEG signals are enhanced by the CSP algorithm for channel reduction and separability enhancement. Second, the temporal parameters of AR model and the CSP-log algorithm obtained from each row of the enhanced signals compose a feature vector where each feature stands for a power sample in frequency or spatial spectrum. Third, feature selection by the proposed method is employed on all feature vectors of the training trials to select the optimal feature subset through evaluating the discrimination capacity of each individual feature and simultaneously select the optimal subject-specific time segment. In the end, the dimensionality reduced feature vectors in the optimal time segment are employed to train a linear discrimination analysis (LDA) classifier which is most commonly used in EEG signals based motor imagery pattern classification. Afterwards, the correct classification accuracy and the session-to-session transfer kappa coefficient are computed based on the competition rule to validate the superiority of the proposed method. The contributions and novelty of this paper are summarized as follows. Firstly, we propose a novel statistical method to select the discriminative feature subset to address the concerns within filter and wrapper approaches. Secondly, a novel discriminative time segment selection for BCI application based on KL-divergence is developed. Thirdly, the Parzen window is introduced in this paper to estimate the conditional probability for each single feature without requiring permutation over all features. Finally, this paper provides a potential framework for combing other feature extraction models and classification algorithms with the proposed optimal feature and time segment selection scheme for enhancing EEG signals classification performance.

The remainder of this paper is organized as follows. Section 2 presents the feature extraction procedure and the details of the proposed method, including the optimal feature subset selection and the optimal time segment selection. In section 3, we introduce two public datasets used in this study and the details of experimental study, including experiment setup and comparative experiments. The experimental results and conclusion are presented is section 4 and 5, respectively.

2. Method

In this section, we make a brief introduction to two powerful and efficient feature extraction algorithms. Besides, we detail a novel KL-divergence based method to select the optimal feature subset and time segment. The block diagram of the proposed method is illustrated in Fig. 1 in which the details of each part would be discussed below.

From the aforementioned, a classical BCI system involves four indispensable steps: the acquisition of multichannel EEG signals, signal preprocessing, feature extraction and EEG signals based motor imagery pattern classification. Fig. 2 gives the reference plots of three channels from the first subject of a dataset which will be thoroughly discussed in section 3. It could be seen that the EEG signals are totally buried in noise, which means the preprocessing and discriminative feature selection are very important in EEG signals classification. Besides, the details of signal acquisition and preprocessing, including artifact detection, noise elimination and bandpass filtering will not be discussed for the sake of clarity since the same procedures have been adopted as other BCI systems.

2.1. Feature extraction

2.1.1. Common spatial pattern for enhancement and feature extraction

Common spatial pattern (CSP) is a spatial optimization algorithm, which is designed to extract spatial filters to encode the most discriminative information and first introduced in Ref. [16]. It aims to obtain the optimal spatial projections that maximize the variance ratio for two

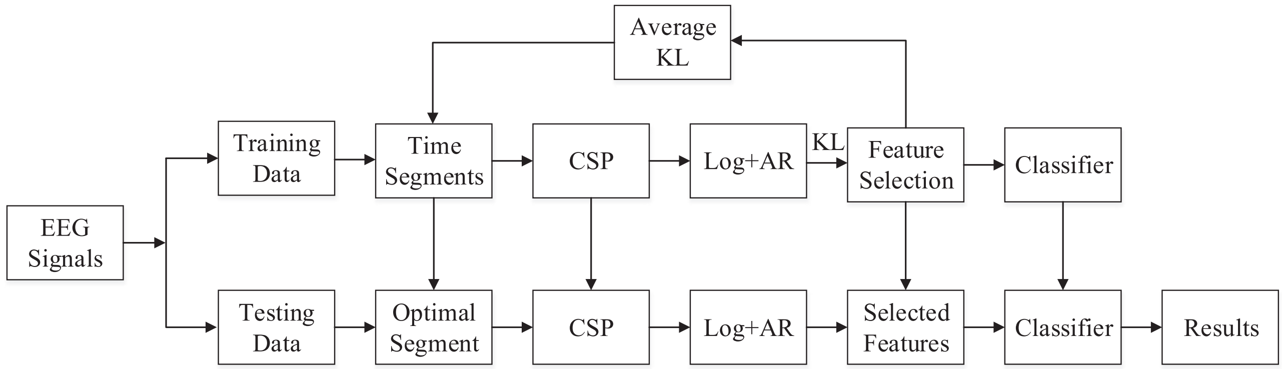

Fig. 1. The block diagram of our KL-divergence based optimal feature and time segment selection method for the training and testing phases.

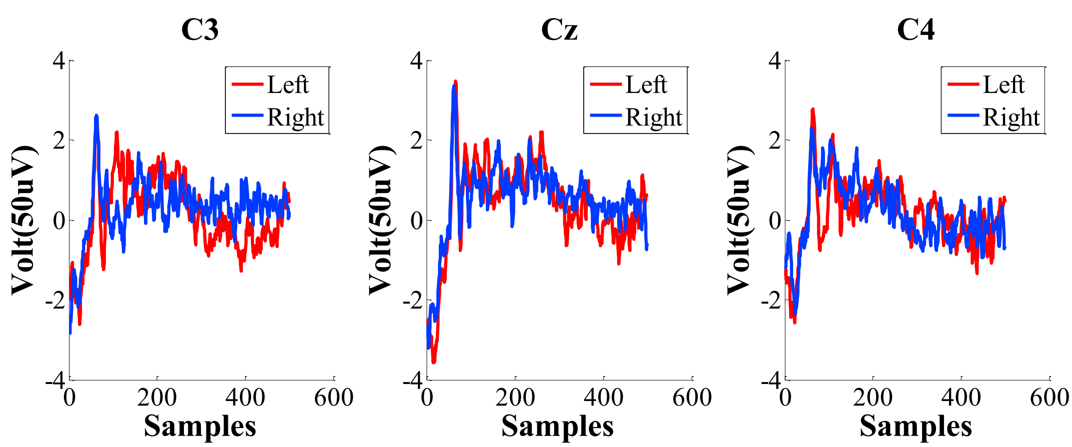

Fig. 2. The reference plots of 3 channels (C3, Cz and C4) for the EEG signals from the first subject of dataset 2b.

motor imagery patterns. The effectiveness of CSP deeply depends on the predetermined time segment. In this study, a series of 2s time segment EEG signals $\mathbf{X} \in R^{C\times T}$ corresponding to two patterns (left hand and right hand) are employed to construct a projection matrix, where C is the number of channels and T is the number of time points of each channel in 2s segment. Generally, the first and last m columns of the projection matrix are retained to form a new matrix $\mathbf{W} \in R^{C\times 2m}$. In classical CSP enhancement algorithm, an input EEG matrix $\mathbf{X}$ is transformed as follows

$$\mathbf{Y} = \mathbf{W}^{\mathrm{T}}\mathbf{X} \tag{1}$$

CSP can be considered as a signal enhancement algorithm [4], which enhances the class separability measure. Accordingly, we expect to extract feature vectors with more class separability information after CSP. Besides, CSP-log itself is also a powerful and efficient feature extraction algorithm. An enhanced EEG matrix $\mathbf{Y}$ could be employed to construct a $2m$-dimensional feature vector in which each feature is calculated as the variance as follows

$$f_q = \log\left(\frac{var(\mathbf{Y}_q)}{\sum_{q=1}^{2m} \mathrm{var}(\mathbf{Y}_q)}\right) \tag{2}$$

the symbol f_q denotes the q-th element of a feature vector.

2.1.2. Autoregressive model for feature extraction

The autoregressive (AR) model is a pure mathematical model to detect the activity changes of EEG signals [18,19], which is able to describe completely the second order statistics of a time series. It has proved that the AR model is a maximum entropy spectrum estimation, which accurately describes the spectral density function using a minimum number of parameters [20,21]. The AR model specifies that each output variable could be predicted by a linear combination of the previous p values where the linear combination weights would be considered as AR coefficients. The AR model is calculated as follows

$$y_{i,k} = a_{i,1}y_{i,k-1} + a_{i,2}y_{i,k-2} + \cdots + a_{i,p}y_{i,k-p} + e_k \tag{3}$$

Therefore, an AR model involves the calculation of p coefficients, which could be employed to predict y_{ik} at time k of the $i-$th row of matrix $\mathbf{Y}$ by employing the linearly weighted combination of coefficients $a_{i,j}$ $(j = 1, 2, 3, \cdots, p)$ with the p (model order) previous variables. The symbol e_k is a random noise perturbation at time$k$. By taking the z-transformation of equation (3), we can obtain

$$H(z) = \frac{Y(z)}{E(z)} = \frac{1}{1 - \sum_{n=1}^{p} a_{i,n}z^{-n}} \tag{4}$$

The power spectrum of the EEG signals can be obtained by evaluating $H(z)$ on the unit circle where $z = \exp(jw)$ as follows

$$P(f) = \frac{P(p)\cdot \Delta t}{\left|1 - \sum_{n=1}^{p} a_{i,n}\exp(-2\pi jfn\Delta t)\right|} \tag{5}$$

In equation (5), $P(p)$ is the mean output power and Δt is the sampling time interval. From the aforementioned, the AR model can be employed to estimate the power spectrum of EEG signals. Hence, the AR coefficients could be employed as the features to detect the power spectrum of EEG signals. In this study, the model order p has been set to 4, and the coefficients are obtained by the Yule-Walk method [22]. It should be stressed that the use of larger model orders does not entail significant improvement but results in a higher dimensionality of feature space [23]. Furthermore, the AR model would provide higher resolution spectral estimations from short EEG intervals while 1s is sufficient to produce a spectrum with a well-defined peak [24]. Therefore, short time segment AR model usually performs better in EEG feature extraction. However, in the enhancement procedure, the EEG signals are processed in the 2s time interval for the convenience of CSP method. Accordingly, we can divide the enhanced signals into two none overlapping 1s time interval signals in which AR model is employed to extract the same number coefficients as features, respectively. Combining AR model with CSP-log method together, this feature extraction scheme results in $2m(1+2p)$ features in total for one EEG trial in which $2m$ features correspond to the CSP-log algorithm and $4mp$ features correspond to two short AR models, respectively.

2.2. Feature selection

Feature selection is widely adopted for reducing the dimensionality of feature space in many areas. In this study, we propose a novel KL-divergence method based feature selection algorithm to rank the original features according to their discriminative capacities, which retains the advantages involved in the filter approaches and the wrapper approaches. The proposed method is able to select those discriminative features based on the MAP rule. From the aforementioned, after the feature extraction procedure, we can extract one feature vector $\mathbf{V_i} \in R^{1\times 2m(1+2p)}$ from the $i-$ th training trial. Here, we assume that the feature matrix $\mathbf{F} \in R^{2m(1+2p)\times nt}$ is composed of all training feature vectors.

$$\mathbf{F} = \left[\mathbf{V}_1^T, \mathbf{V}_2^T, \mathbf{V}_3^T, \cdots, \mathbf{V_{nt}^T}\right]^{\mathbf{T}} = \left[\mathbf{f}_1^T, \mathbf{f}_2^T, \mathbf{f}_3^T, \cdots, \mathbf{f_N^T}\right] \tag{6}$$

where nt is the number of training trials and $N=2m(1+2p)$ is the dimensionality of the original feature space. The symbol $\mathbf{f_i^T} \in R^{nt\times 1}$ $(i = 1,2,3,\cdots,N)$ denotes the $i-$ th feature of the feature matrix. The true class labels are denoted as follows

$$\begin{aligned}&\mathbf{\Omega} = [\Omega_1, \Omega_2, \Omega_3, \cdots, \Omega_{nt}]^{\mathbf{T}}\\&\Omega_j \in \{\omega_1, \omega_2\}\end{aligned} \tag{7}$$

where Ω_j denotes the label of $j-$ th training trial. We use ω_1 and ω_2 to denote the left hand and right hand motor imagery patterns, respectively. Feature selection in this paper is designed to select M features from the original N features, and the selected features construct a feature subset which will be employed to train a classifier and predict a new testing feature vector. We assume that each class is represented as a particular distribution, P_1 for class ω_1 and P_2 for class ω_2 as follows

$$\begin{aligned}P_1 &= p(f|\omega_1)\\P_2 &= p(f|\omega_2)\end{aligned} \tag{8}$$

where $p(f|\omega_i)$ denotes the class conditional probability distribution in which f is a feature of matrix $\mathbf{F}$. Therefore, this classification procedure can be regarded as a binary hypothesis testing. Hence, if a feature is drawn from P_1, we accept the hypothesis that we should attribute this feature to class ω_1. According to the information theory [13,25], the Kullback-Leibler divergence of a feature between two probability distributions (P_1 against P_2) can be defined as follows

$$\begin{aligned}KL(P_1,P_2) &= \int_f p(f|\omega_1) log\frac{p(f|\omega_1)}{p(f|\omega_2)}df\\&= E_{p1}\left[log\frac{p(f|\omega_1)}{p(f|\omega_2)}\right]\end{aligned} \tag{9}$$

From the definition of this model, the KL-divergence measure presents the mean posteriori information of likelihood ratio for discrimination in favor of ω_1 against ω_2. Under the MAP rule, we will classify the feature f into class ω_1 (acceptP_1), if

$$\log\frac{P(f|\omega_1)}{P(f|\omega_2)} > -\log\frac{P(\omega_1)}{P(\omega_2)} \tag{10}$$

Hence, the KL-divergence measure between two probability distributions is an indicator of discriminative capacity, which is associated with the MAP rule. Especially, the KL-divergence measure of a feature between two discrete distributions is given by

$$KL\left(P_1^i, P_2^i\right) = \sum_{j=1}^{nt} p\left(f_{i,j}|\omega_1\right)\log\frac{p\left(f_{i,j}|\omega_1\right)}{p\left(f_{i,j}|\omega_2\right)} \tag{11}$$

where $f_{i,j}$ denotes the $j-$ th value of $i-$ th column of feature matrix $\mathbf{F}$. From the extension of the central limit theorem, it has been shown that the first type error α^*, which is the probability of incorrectly accepting P_1 for a large number of training trials, could be calculated asymptotically [26] as

$$\log\frac{1}{\alpha^*} = \lim_{nt\to\infty} KL(P_1, P_2; nt) \tag{12}$$

Accordingly, a larger value of divergence measure of ω_1 against ω_2 presents a lower first type error when we have infinite number of trials. It also should be noticed that the divergence measure here is not symmetric.

$$KL(P_1, P_2) \neq KL(P_2, P_1) \tag{13}$$

Alternatively, the KL-divergence measure also presents the discriminative capacity of a feature in favor of ω_2 against ω_1. Similarly, the second type error β^*, which is the probability of incorrectly accepting P_2 for a large number of training trials, could be calculated asymptotically [26] as

$$\log\frac{1}{\beta^*} = \lim_{nt\to\infty} KL(P_2, P_1; nt) \tag{14}$$

Accordingly, a larger value of divergence measure of ω_2 against ω_1 presents a lower second type error when we have infinite number of trials. To take into account both two type errors in an asymptotic way, the symmetric KL-divergence of a feature can be redefined as

$$KL^i = KL\left(P_1^i, P_2^i\right) + KL\left(P_2^i, P_1^i\right) \tag{15}$$

A larger KL value would contribute to a smaller two type errors asymptotically. Therefore, the KL-divergence in this way is able to estimate the difficulty and capacity of discriminating between two distributions. Hence, we can expect more discriminative capacity from those features with larger KL values. In this paper, it is natural to select those features with maximum discriminative capacities by maximizing the KL-divergence. However, there are still some concerns in this model. The main problem involved in this paper is the calculation of conditional probability of $f_{i,j}$ given class Ω. As discussed in the aforementioned, the evaluation of KL-divergence is for a single feature where we calculate the KL-divergence of each feature one by one. It is an efficient method that could be employed to estimate the conditional probability within a single feature based on Parzen window when we do not have any prior distribution information, which is introduced in Refs. [15,27]. In this paper, the conditional probability of a single feature can be estimated as follows

$$p\left(f_{i,j}|\Omega\right) = \frac{1}{n_\Omega}\sum_{r\in I_\Omega}\phi\left(f_{i,j} - f_{i,r}, h\right) \tag{16}$$

where n_Ω is the number of trials in the training data corresponding to class Ω, and I_Ω is the set of indices of the training trials corresponding to class Ω. The symbol ϕ is a smoothing kernel function of vector $\mathbf{y^T}$ and smoothing parameter h.

$$\phi(\mathbf{y}^{\mathrm{T}}, h) = \frac{1}{\sqrt{2\pi}} \exp\left(\frac{-\left\| \mathbf{y}^{\mathrm{T}} \right\|^{2}}{2h^{2}} \right) \tag{17}$$

It is a univariate Gaussian kernel function, which is widely used in statistics and machine learning. The smoothing parameter h can be calculated [28] as follows

$$h = \left(\frac{4}{3 n_{\Omega}} \right)^{\frac{1}{5}} \sigma(\mathbf{y}^{\mathrm{T}}) \tag{18}$$

where σ denotes the standard deviation function of vector $\mathbf{y}^{\mathrm{T}}$ from (17).

Based on the above calculation model, the KL-divergence value of each individual feature can be computed. Then, sort all the features in descending order. In this step, we rank the KL-divergence over all features, which is given by

$$KL^{e_1} \geq KL^{e_2} \geq \cdots \geq KL^{e_M} \geq \cdots \geq KL^{e_N} \tag{19}$$

where e_i denotes the feature index. Accordingly, this efficient method ranks the importance of each individual feature by estimating its discriminative capacity when only one single feature is used. The first indexed M features that possess the largest KL-divergence values are expected to possess the best discriminative capacity. The top ranked M features are selected to construct the optimal feature subset S_M as follows

$$S_M = \left\{ \mathbf{f}_{e_1}, \mathbf{f}_{e_2}, \cdots, \mathbf{f}_{e_M} \right\} \tag{20}$$

The proposed approach in this way is much more efficient as the discriminative capacity of each individual feature is computed once, and it has the computation complexity of $O(M)$.

2.3. Optimal time segment

Generally, the feature extraction is performed in a fixed 2s time segment, such as the widely employed 0.5–2.5s from the onset of the visual cue presented for the subjects to perform motor imagery patterns, which is the same as the entry of the competition [29]. Previous experimental result had shown that the effectiveness of feature extraction depended on the subject-specific time segment of the EEG signals taken relative to the visual cue [30]. But it is often selected manually or heuristically [31]. To address this issue, the KL-divergence based feature selection scheme supplies an appropriate approach to automatically select the optimal time segment. In order to illustrate this case, we extract features from 21 overlapping time segments which are: 0–2, 0.1–2.1, 0.2–2.2$\cdots$, 2–4s from the onset of the visual cue presented for the subjects to perform motor imagery patterns. In this way, all the time segments include a length of 4s EEG signals, which is in strict accordance with the practical BCI datasets.

After the calculation of KL-divergence and feature ranking over all these 21 overlapping time segments, the average KL-divergence of the selected features is calculated as follows

$$KL(S_M) = \frac{\sum\limits_{\mathbf{f}_i \in S_M} KL^{i}\left(P_1^{i}, P_2^{i}\right)}{|S_M|} \tag{21}$$

The optimal time segment that contains the most discriminative information is selected as the one that has the maximal average KL-divergence value. Fig. 3 presents the average KL-divergence values of 4 top ranked features over all 21 overlapping time segments for 6 subjects. Fig. 3 demonstrates that the selection of optimal subject-specific time

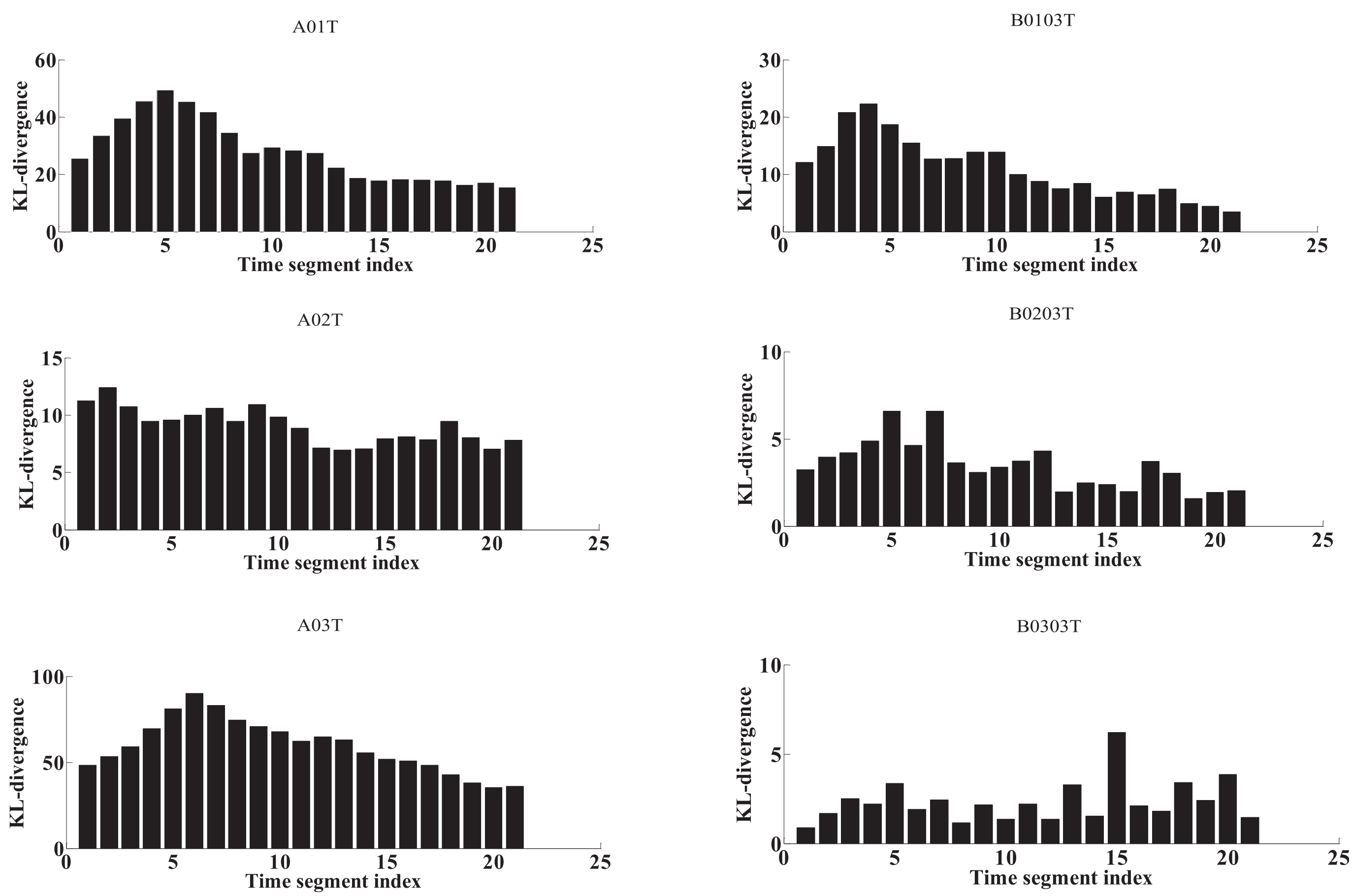

Fig. 3. The average KL-divergence of the 4 top ranked features calculated over all 21 time segments based on the training data. The first column corresponds to three subjects (Subject1, Subject2 and Subject3) of dataset 2a and the second column corresponds to three subjects (Subject1, Subject2 and Subject3) of dataset 2b. The time segment index indicated as 1, 2, 3, …, 21 correspond to the time segments 0–2s, 0.1–2.1s, 0.2–2.2s, …, 2–4s, respectively.

segment is helpful to extract the features with more discriminative information based on the proposed method.

3. Experimental study

3.1. Experimental datasets

The experiments are performed using two public datasets, BCI Competition IV dataset 2a and dataset 2b, which comprise the training and evaluation data from 18 subjects. For dataset 2a, 9 subjects are asked to perform four different motor imagery patterns that are composed of left hand, right hand, foot and tongue. The training and evaluation data of each subject consists of one training session and one evaluation session of single-trial EEG signals, respectively. For each subject, there are 72 trials in training session and 72 trials in evaluation session for each pattern recorded from 22 channels with a sampling frequency of 250Hz. The visual cue indicating the motor imagery pattern in each trial lasts for 4s. The details of this dataset can be seen in Ref. [32]. As the binary classification is predominately employed in basic BCI design. Only the trials corresponding to the left hand and right hand movements are selected for this presented study. For dataset 2b, 9 subjects are asked to perform two different motor imagery patterns that are composed of left hand and right hand. The training and evaluation data of each subject consists of three training sessions and two evaluation sessions of single-trial EEG signals, respectively. However, only the 3rd training session is employed for training because its feedback scheme is in accordance with the two evaluation sessions. Hence, for each subject, there are 60 trials in training session and 120 trials in evaluation sessions for each motor imagery pattern recorded from 3 channels with a sampling frequency of 250Hz. The visual cue indicating the motor imagery patterns in each trial lasts for 4.5s. It should be noticed that the choice of m filters in CSP is set to 3 for dataset 2a and 1 for dataset 2b. The former is selected as recommended in Ref. [33], because a great choice of m would not significantly improve classification accuracy. The latter is selected because there are only three channels available.

3.2. Experimental setup

The proposed optimal feature subset and time segment selection method based on the KL-divergence is extensively evaluated in four steps. First, feature extraction is performed in the widely used 0.5–2.5s time segment and the correct classification accuracy is measured by selecting different number of features ranged from 1 to all (54 for dataset 2a and 18 for dataset 2b). Second, given a minimum subset of M from the original set of N features, the KL-divergence of each single feature and the average KL-divergence of the top ranked features are calculated over all time segments. The optimal time segment is the one with the largest average KL-divergence value. Therefore, the optimal time segments could be determined automatically when the dimensionality of feature subset is given, simultaneously. 10-fold cross-validation scheme is employed on the feature matrices from training datasets 2a and 2b to choose the dimensionality of the optimal feature subset. The cross-validation classification accuracy by LDA classifier is considered as the benchmark to determine the optimal feature subset and time segment. Third, we could obtain the final session-to-session transfer classification accuracy of evaluation data based on the optimal feature subset and time segment. It should be stressed that the session-to-session transfer performance is more challenging in practical BCI because signals may change substantially from the training data to the evaluation data since they are recorded from separate days [34,35]. In the final step, kappa coefficient, evaluated on the entire single-trial EEG signals from the onset of the visual cue, is employed as the performance measure in this study. It normalizes the distribution of wrong classifications and the frequency of occurrence, which can be calculated in Refs. [36,37] as follows

$$kappa = \frac{P_a - P_c}{1 - P_c} \tag{22}$$

where P_a is the proportion of agreement, which is the correct classification accuracy, and P_c is the proportion of chance agreement, which is equal to the accuracy of a trivial or random classifier. Based on the proposed method, kappa value is computed from the onset of the fixation cross to the end of the cue for every time point across all the evaluation trials. It should be stressed that the maximum kappa efficient is the evaluation principle in the competition [32].

To evaluate the method, some other advanced and competitive algorithms are introduced. First, as discussed in the first step, experiments should be performed to evaluate the efficiency of our KL-divergence based feature selection scheme in comparison with others. As discussed in the aforementioned, filter approaches are predominately employed in the feature selection for its simplicity and efficiency. Much more computation expense involved in wrapper approaches makes them unacceptable in practical BCI. Therefore, the compared algorithms discussed in the paper are mainly attributed to the filter approaches. Principle component analysis (PCA) is a classical and mathematical procedure that an orthogonal transformation is used to concert a set of feature vectors into a set of values of linearly uncorrelated feature vectors in which those variables possess largest variance are called principle components. It is a powerful approach to reduce the dimensionality of feature space in EEG based motor imagery patterns classification. Another mathematical procedure evaluates the area between the empirical receiver operating characteristic curve (ROC) and the random classifier slop to select features [38]. It is employed for each single feature to estimate the accuracy to error ratio based on posterior probability between patterns. Different from the ROC algorithm, the proposed method is employed for each single feature to estimate the divergence based on entropy between patterns. Another entropy based filter approach termed fast correlation-based filter solution (RCBF) is also proposed in feature selection area [39], which characterizes the purity of an arbitrary collection of examples [40]. Based on the entropy based methods, FBCSP achieves a great success in EEG signals classification [41], which decomposes a time segment EEG signals into series bandpass filtered signals. Then, CSP as the feature extraction is employed separately to extract feature vectors and mutual information is used to select the target relevant features. Cross-validation (CV) scheme is a classical feature selection approach, which ranks features by their cross-validation accuracy even though classification algorithms are involved. Similar to CV scheme, support vector machine recursive feature elimination (RFE) select those features that contribute to classification most by training a classifier [42]. A latest algorithm termed dynamic frequency feature selection (DFFS) approach is proposed to select the most discriminative features [43]. It selects the optimal features one by one based on their error performance without omitting those selected before. Therefore, a drawback involved in this algorithm is that the most discriminative features would be selected more than one time. Besides, additional experiments are performed to demonstrate that the optimal time segment selection scheme contributes to improve the classification performance. We perform experiments on feature selection versus all features and optimal time segment selection versus the fixed time segment (0.5–2.5s).

4. Experimental results

In this section, the experimental results would be presented. Based on the same original feature set, we compare the proposed KL-divergence based feature selection approach (KL) with other advanced feature selection methods, including the principle component analysis (PCA), receiver operating characteristic (ROC), fast correlation–based filter (FCBF), cross-validation (CV), support vector machine recursive feature elimination (RFE) and dynamic frequency feature selection (DFFS). We carry out experiments on these six benchmarks to present the superiority of the proposed feature selection scheme.

Fig. 4 presents the classification performance obtained by the proposed feature selection scheme and six benchmarks when different number of features are selected. We present the classification accuracy curves of 6 subjects (3 from dataset 2a and 3 from dataset 2b) when different number of features are selected, and the number of selected features ranges from 1 to almost half of original features (20 for dataset 2a and 10 for dataset 2b). On one hand, feature selection aims to search an optimal feature subset from the original set, the fewer of features the better. On the other hand, the details of classification performance could be clearly presented by fewer features. We should have presented the former 3 subjects for each dataset, respectively. However, subject 2 of 2a and subject 3 of 2b present a totally irregular trend and poor classification performance for all feature selection benchmarks, and the distinctions among these feature selection methods cannot be measured by visible inspection.

It can be clearly shown that the classification performance may not be improved when more features are selected. The optimal feature subset is just composed of several elements from the original feature set. Therefore, selecting the most discriminative features is of great significance to lowering the computation cost and improving the classification performance, which is meaningful in practice. Further, on one hand, the proposed KL-divergence method performs better than others in most cases; on the other hand, it has the potential to obtain better classification performance by fewer number of features. For instance, only by selecting 2 top ranked features can the subject 1 of dataset 2a achieve the highest classification accuracy. From the six subfigures, there are no significance difference among FCBF, CV, ROC and DFFS methods. Meanwhile, FCBF, CV, ROC and DFFS methods perform better in comparison with PCA, but they are inferior to the proposed method. The SVM-REF method presents a relatively better feature selection performance for subjects of dataset 2b but presents a relatively poor feature selection performance for subjects of dataset 2a. In a word, comparisons between the KL-divergence based feature selection and these benchmarks on two public datasets have been conducted, which have shown the superiority of the proposed method in the field of feature selection.

Tables 1 and 2 present the experimental results on the correct classification accuracy from the training sessions to the evaluation sessions of dataset 2a and 2b, respectively. The second column termed proposed method presents the classification accuracy of every subject obtained by the combination of KL-divergence based feature selection and optimal time segment selection. The optimal number of selected features is determined by 10-fold cross-validation scheme, and the time segment with maximum average KL-divergence would be selected automatically. To verify whether the feature and time segment selection by KL-divergence could improve the classification performance, comparison experiments with and without feature and time segment selection have to be conducted. We can conclude from the results listed in Tables 1 and 2 that the proposed method yields the highest averaged classification accuracy of 81.99% for dataset 2a and 77.22% for dataset 2b, respectively. Compared with all features in the fixed time segment scheme (AFFS), feature selection in the fixed time segment scheme (FSFS) achieves better classification performance with an average improvement of 10.05% for dataset 2a and 3.46% for dataset 2b. Accordingly, feature selection by KL-divergence achieves a great success for classification especially for dataset 2a. Besides, compared with AFFS scheme, all features in the optimal time segment scheme (AFOS) also achieves better classification performance with an average improvement of 6.49% for dataset 2a and 1.22% for dataset 2b. Especially, after combining the feature and optimal time segment selection together, we can obtain an obvious significant improvement with an average improvement of 13.32% for 2a and 4.52% for 2b, respectively. Further, the benchmarks introduced above are also included to compare against the proposed method. These benchmarks except PCA outperform AFFS almost for all subjects but are inferior to FSFS and the proposed method, which demonstrate that the classification performance would benefit from appropriate feature selection schemes and the proposed scheme is superior to any benchmark, especially when the optimal time segment selection scheme is employed together.

Table 3 presents the experimental results on the kappa coefficient of the session-to-session transfer from the training session to the evaluation sessions of dataset 2b. In this table, the kappa coefficient as a normalized criterion shows all competitive methods significantly outperform a

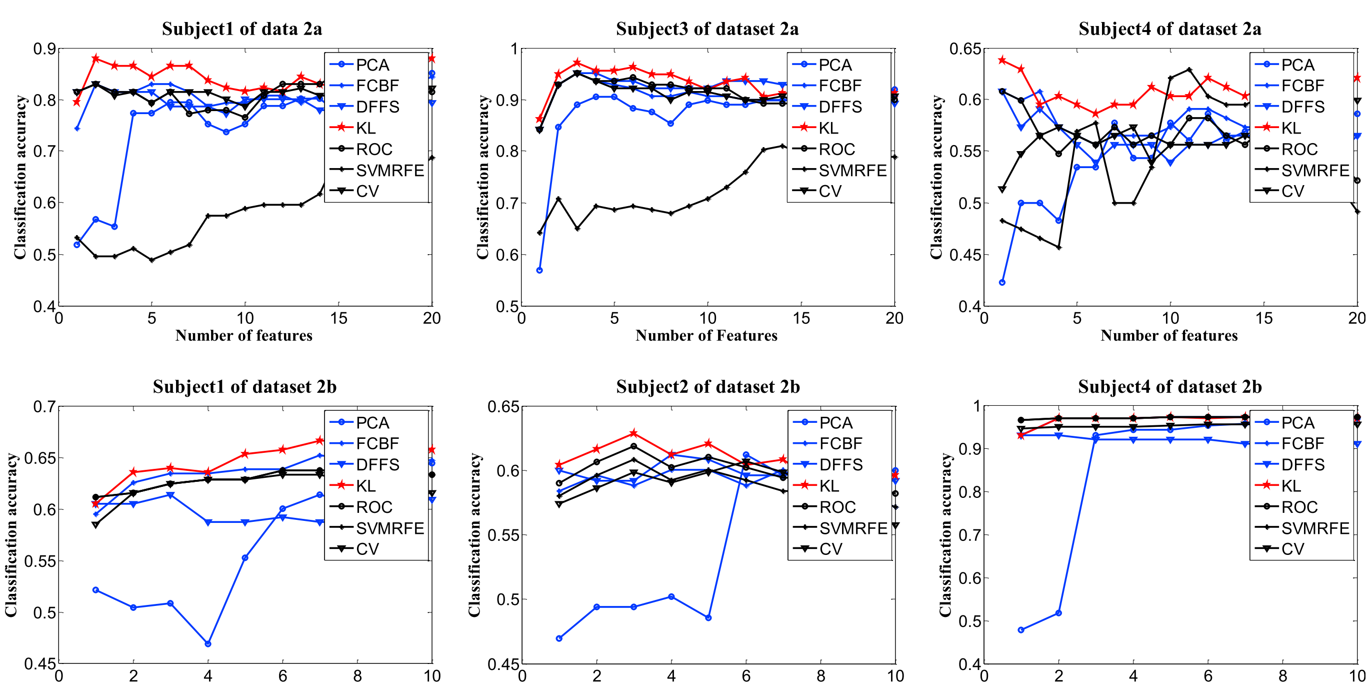

Fig. 4. The classification accuracy obtained by different feature selection methods. X-axis presents the number of features we select based on different feature selection methods. Y-axis presents the classification accuracy obtained from the selected features. The first row corresponds to three subjects from dataset 2a and the second row corresponds to three subjects from dataset 2b, respectively.

Table 1
Experimental results on the correct classification accuracy (%) on dataset 2a performed using the proposed method against feature selection in the fixed time segment (FSFS), all features in the optimal time segment (AFOS) and all features in the fixed time segment (AFFS). Five well performed feature selection methods, FCBF, DFFS, ROC, CV and PCA, are also included.

Subject	Proposed method	FSFS	AFOS	AFFS	FCBF	DFFS	ROC	CV	PCA
A01	0.8794	0.8794	0.7872	0.7872	0.8652	0.8652	0.8652	0.8652	0.7730
A02	0.6549	0.6197	0.5634	0.5352	0.5563	0.5634	0.5915	0.5704	0.5211
A03	0.9708	0.9708	0.9197	0.8759	0.9562	0.9708	0.9562	0.9562	0.9051
A04	0.7500	0.6379	0.7414	0.4914	0.6034	0.6034	0.5776	0.5776	0.4828
A05	0.6370	0.6148	0.5704	0.5704	0.5407	0.5111	0.6296	0.4963	0.5037
A06	0.7500	0.6759	0.6296	0.6019	0.6944	0.6759	0.6944	0.5648	0.4167
A07	0.8286	0.7786	0.7357	0.6929	0.7714	0.7571	0.7714	0.7286	0.4857
A08	0.9851	0.9851	0.9403	0.9104	0.9627	0.9328	0.9627	0.9627	0.9254
A09	0.9231	0.9231	0.8769	0.7154	0.9154	0.9231	0.9154	0.9154	0.9077
Average	0.8199	0.7872	0.7516	0.6867	0.7629	0.7559	0.7738	0.7375	0.6579

Table 2
Experimental results on the correct classification accuracy (%) on dataset 2b performed using the proposed method against feature selection in the fixed time segment (FSFS), all features in the optimal time segment (AFOS), and all features in the fixed time segment (AFFS). Five well performed feature selection methods, FCBF, DFFS, ROC, CV and RFE, are also included.

Subject	Proposed method	FSFS	AFOS	AFFS	FCBF	DFFS	ROC	CV	RFE
B01	0.6711	0.6667	0.6535	0.6228	0.6447	0.5877	0.6491	0.6360	0.6491
B02	0.6082	0.5959	0.5959	0.5755	0.6204	0.6122	0.6122	0.6204	0.6367
B03	0.5348	0.5522	0.5043	0.5217	0.5174	0.4826	0.4957	0.5217	0.4957
B04	0.9739	0.9674	0.9709	0.9642	0.9707	0.9218	0.9707	0.9707	0.9739
B05	0.8242	0.7802	0.8022	0.7802	0.7399	0.7179	0.7143	0.7363	0.7143
B06	0.7809	0.7649	0.7291	0.6932	0.7689	0.7610	0.7729	0.7649	0.7729
B07	0.8060	0.7974	0.7672	0.7672	0.7543	0.7414	0.7543	0.7543	0.7543
B08	0.9261	0.9217	0.8913	0.8913	0.9174	0.8870	0.9174	0.9174	0.9174
B09	0.8245	0.8082	0.7388	0.7265	0.7714	0.7510	0.7714	0.7714	0.7714
Average	0.7722	0.7616	0.7392	0.7270	0.7450	0.7181	0.7398	0.7437	0.7428

Table 3
Experimental results on the kappa coefficient on dataset 2b performed using the proposed method against other latest competitive methods, including OSTP, FBCSP, ECBF, DFFS, ROC, CV, RFE and 2 winners in the competition.

Subject	Proposed method	OSTP	FBCSP	FCBF	DFFS	ROC	CV	RFE	1st	2nd
B01	0.43	0.43	0.36	0.34	0.25	0.34	0.36	0.28	0.40	0.42
B02	0.34	0.21	0.17	0.30	0.22	0.30	0.30	0.31	0.21	0.21
B03	0.23	0.24	0.17	0.22	0.08	0.18	0.19	0.20	0.22	0.14
B04	0.95	0.94	0.96	0.95	0.79	0.95	0.95	0.95	0.95	0.94
B05	0.78	0.84	0.85	0.68	0.30	0.65	0.68	0.68	0.86	0.71
B06	0.61	0.59	0.59	0.58	0.40	0.60	0.59	0.60	0.61	0.62
B07	0.70	0.58	0.56	0.66	0.61	0.65	0.66	0.66	0.56	0.61
B08	0.87	0.86	0.86	0.86	0.69	0.87	0.86	0.87	0.85	0.84
B09	0.69	0.66	0.75	0.65	0.38	0.66	0.66	0.66	0.74	0.78
Average	0.62	0.59	0.59	0.58	0.41	0.58	0.58	0.58	0.60	0.58

chance classification. The results also present the proposed method yields the largest averaged kappa coefficient across all subjects compared to other advanced methods. In the final experiments, the optimal number of features is the same as how many we select before by 10-fold cross-validation.

The mutual information-based selection of optimal spatial-temporal patterns (OSTP) method [35] simultaneously optimize spatial and temporal patterns by optimizing the feature selection based on mutual information entropy and selecting the optimal time segment. Its authors show that the OSTP method is superior to their previous work FBCSP [41] which only optimizes spatial pattern and selects the optimal time segment manually, even though they have the similar classification performance for dataset 2b [35]. It should be noted that their previous work FBCSP on the evaluation data achieved the best averaged kappa coefficient among all submission of the competition [29]. OSTP and FBCSP perform CSP on various bandpass filtered EEG signals and obtain a larger number of features but select a fixed number of them. The proposed method performs CSP and AR on one wide bandpass filtered EEG signals and obtain a fewer number of features but select an optimal feature subset automatically. Besides, the proposed method makes an obvious difference in comparison with OSTP and FBCSP, especially for subject2 and subject7. Meanwhile, the proposed method outperforms other feature selection method and the entry of this competition in kappa coefficient criterion.

5. Conclusion

In this paper, we present a novel statistical model based on KL-divergence to select the optimal feature subset and the optimal subject-specific time segment in the EEG signals based motor imagery patterns classification. At the beginning of our experiments, we perform the common spatial pattern algorithm on a time segment of 7–30Hz band-pass filtered EEG signals, since it enhances class separability between different motor imagery patterns. This step would be of benefit in extracting more discriminative feature vectors. In the feature extraction procedure, the autoregressive model and log-variance are employed on the CSP enhanced EEG signals. These extracted elements are spatial and temporal correlated power spectral features. When feature number is determined, we can select the optimal feature subset and optimal time segment simultaneously. The optimal feature number is determined by cross-validation scheme on training data. Based on this measure, the correct classification accuracy results are obtained, and two public datasets validate that the proposed method achieves a great success for classification performance with an average improvement of 13.32% for 2a and 4.52% for 2b, respectively. In the end, the kappa coefficient is also employed for dataset 2b to demonstrate that the proposed method achieves the best mean kappa coefficient among submissions for the competition and many state-of-the-art methods. Besides, to validate the proposed technique for feature selection would perform well with other classifiers, we carry out some experiments based on the proposed method with support vector machine (SVM) and Naïve Bayesian classifiers. Experimental results indicate that both of them have the similar classification accuracy and kappa coefficient with LDA classifier. SVM classifier yields an averaged classification accuracy of 81.99% for dataset 2a, 76.10% for dataset 2b and kappa coefficient of 0.61. Naïve Bayesian classifier yields an averaged classification accuracy of 80.13% for dataset 2a, 79.89% for dataset 2b and kappa coefficient of 0.62. On one hand, LDA is a very simple classifier so that we can train it in less time in comparison with other advanced classifiers; on the other hand, it is more robust without thinking about the distribution of features. Therefore, we choose the LDA in our study and present the details of experimental results for each subject.

However, the proposed method still involves some concerns even though it yields better classification performance than many mainstream feature selection schemes and state-of-the-art methods. The enhancement algorithm CSP is only effective for binary classification although one versus the rest (OVR) has been developed for overcoming its drawbacks. The OVR scheme involves more classifiers to determine one pattern, but sometimes the results of all classifiers cannot obtain a unique and accurate conclusion. Besides, the KL-divergence based feature selection scheme still has some multi-class extension concerns involved. It should be stressed that, to fairly compare the proposed method with the state-of-the-art feature selection methods, two classical and standard feature extraction algorithms are employed as the basic of our feature subset and time segment selection approach. In recent years, feature learning has been a novel direction for feature extraction and classification [44]. Feature learning, or representation learning, aims to learn transformation when building classifiers or other predictors. A good representation is often the one that captures the posterior distribution of the underlying explanatory factor for the observed input. Therefore, feature learning was mainly used for extracting discriminative features. We would expect to extract better features by this new technology to further improve the classification performance. These problems will obtain further exploration in our future study. Nevertheless, the proposed method provides a potential framework for using this statistical feature and time segment selection method for EEG signals decoding and developing other combinations of feature extraction schemes and classification algorithms in the future.

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